NNN-B-00789a (GSA-FSS)

May 19, 1965
Interim Revision of
Fed. Spec. NNN-B-789
March 25, 1963

INTERIM FEDERAL SPECIFICATION

BURET, STRAIGHT, PRECISION

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D.C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification NNN-B-789, dated March 25, 1963.

1. SCOPE AND CLASSIFICATION

1.1 Scope.--This specification covers glass burets suitable for laboratory purposes (see 6.1).

1.2 Classification.

1.2.1 Types, styles, classes, and sizes.--Burets covered by this specification shall be of the following types, styles, classes, and sizes, as specified (see 6.2):

1.2.1.1 Types.

Type I--Borosilicate glass.
Type II --Soda-lime glass.

1.2.1.2 Styles.

Style I--Straight.
Style II--Straight, 3-way stopcock.
Style III--Straight, 3-way stopcock, automatic zero.

1.2.1.3 Classes.

Class A--Precision.
Class B--General Purpose.

1.2.1.4 Sizes.

Size 1A - 5 milliliter (ml), 0.01 ml divisions.	Class A, styles I and II.
Size 1B - 5 ml, 0.02 ml divisions.	Class B, styles I and II.
Size 2A - 10 ml, 0.02 ml divisions.	Class A, styles I and II.
Size 2B - 10 ml, 0.05 ml divisions.	All classes, styles I and II.
Size 3 - 25 ml, 0.1 ml divisions.	All classes, all styles.
Size 4 - 50 ml, 0.1 ml divisions.	All classes, all styles.
Size 5 - 100 ml, 0.2 ml divisions.	All classes, all styles.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.--The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

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Federal Specifications:

DD-G-541--Glass (Laboratory).
UU-T-111--Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566--Boxes, Folding, Paperboard.
PPP-B-601--Boxes, Wood, Cleated-Plywood.
PPP-B-621--Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636--Box, Fiberboard.
PPP-B-676--Boxes, Set-Up, Paperboard.
PPP-C-843--Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102--Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123--Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-M-14077--Molding Plastic, Polytetrafluoroethylene (TFE-Fluorocarbon Resin.)

Military Standards:

MIL-STD-105--Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129--Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications--The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Classification Committee Publication:

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

National Bureau of Standards Publication:

NBS Circular 602 -- Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

Commercial Standard:

CS21 -- Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Borosilicate glass.--Borosilicate glass for type I burets shall conform to the glass requirements of type I, class A of DD-G-541.

3.1.2 Soda-lime glass.--Soda-lime glass for type II burets shall conform to the glass requirements of type II of DD-G-541.

3.1.3 Polytetrafluoroethylene.--Polytetrafluoroethylene shall conform to the composition requirements of class 1 of MIL-M-14077.

3.2 Annealing.--Ring strain shall be no greater than that of Temper No. 1 when tested as specified in 4.5.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.3 Capacity tolerances and dimensions.

3.3.1 Class A.--Class A burets shall have a capacity tolerances in accordance with the requirements of table I when tested as specified in 4.5.2. The dimensions shall be in accordance with table I.

Table I.--Capacity tolerances and dimensions for class A burets.

Size	Capacity of total graduated portion less than and including	Least graduation	Tolerance of total or partial capacity	Scale length	Minimum wall thickness
	ml	ml	ml (±)	cm	mm
1A	5	0.01	0.01	55 ± 5	1.25
2A	10	0.02	0.02	55 ± 5	1.15
2B	10	0.05	0.02	40 ± 5	1.15
3	25	0.1	0.03	40 ± 5	1.1
4	50	0.1	0.05	55 ± 5	1.2
5	100	0.2	0.10	60 ± 5	1.25

3.3.2 Class B.--Class B burets shall have capacity tolerances in accordance with the requirements of table II when tested as specified in 4.5.2.

Table II.--Capacity tolerances and dimensions for class B burets.

Size	Capacity of total graduated portion less than and including	Least graduation	Tolerance of total or partial capacity	Minimum [1] outflow-time	Scale length	Minimum wall thickness
	ml	ml	ml (±)	sec	cm	mm
1B	5	0.02	0.02	40	30 ± 5	1.25
2B	10	0.05	0.04	25	35 ± 10	1.15
3	25	0.1	0.06	30	35 ± 10	1.1
4	50	0.1	0.10	50	55 ± 5	1.2
5	100	0.2	0.20	90	60 ± 5	1.25

[1] Maximum outflow-time shall be 3 minutes.

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3.4 Design.

3.4.1 Shape.--The graduated portion of the buret shall be straight and any cross section of this portion of the buret taken in a plane perpendicular to its longitudinal axis shall be circular. The shape shall permit complete emptying and thorough cleaning. Style II and III burets shall have a filling tube positioned under the stopcock which will permit filling the buret through the stopcock. The filling tube shall be in the form of an elbow, extending downward from the stopcock. The open end of the elbow shall extend toward the rear from the buret and be slightly enlarged and bulbous. At the top, style III burets shall have an ungraduated tapered interior extension terminating in an overflow hole. The exterior of the buret shall extend above the point of overflow of the interior extension and shall have an outlet tube positioned below the overflow point of the interior extension at a point along the junction of the exterior to the interior extension. The end of the outlet tube shall be slightly enlarged and bulbous. The maximum capacity of the interior extension to the initial buret graduation for the 25 and 50 ml burets shall not exceed 1.5 ml and for the 100 ml buret, shall not exceed 3.0 ml.

3.4.2 Stopcocks.--All burets shall have permanently attached stopcocks. With buret in normal viewing position, axis of stopcock shall be perpendicular to plane of ends of short lines with $\not\perp 15^{\circ}$, and handle shall be located on the right. Stopcocks shall not leak when tested as specified in 4.5.3. Stopcock plugs shall be provided with a retaining device and shall be of glass or polytetrafluoroethylene (see 6.2). Glass stopcocks shall conform to Commercial Standard CS21, size 2.

3.4.3 Buret tips.--Buret tips shall be permanently attached to the buret, and shall be made with a gradual taper of 2 to 3 cm, and the end of the tip shall be ground perpendicular to the longitudinal axis of the tip and beveled, or may be firepolished in lieu of grinding. Sudden constriction at the orifice shall not be acceptable. The rate of outflow of burets must be restricted by the size of the tip, and for the graduated length the time of free outflow must not be more than 3 minutes or less than that shown in table III for class A burets and table II for class B burets. For class A burets the limit of error between actual delivery time and marked delivery time shall not exceed that shown in table IV.

Table III. - Minimum delivery times for class A burets.

Length graduated	Time of outflow not less than-	Length Graduated	Time of outflow not less than-
cm	sec	cm	sec
15	30	45	80
20	35	50	90
25	40	55	105
30	50	60	120
35	60	65	140
40	70	70	160

Table IV. - Limits of error in marked delivery time for class A burets

Delivery time (in seconds) less than and including-	Limit of error in marked delivery time
	sec
15	3
20	4
30	6
50	8
100	15
200	20

3.4.4 Dimensions.

3.4.4.1 Class A.--The distance between extreme graduations shall not exceed 70 cm. The lowest graduation shall be at least 2.5 cm above the permanently attached stopcock.

3.4.4.2 Class B.--The dimensions and dimensional tolerances shall be as shown in table II. The lowest graduation shall be at least 2.5 cm above the permanently attached stopcock.

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3.4.4.3 Classes A and B, styles I and II.--The graduation marking for the 5, 10, and 25 ml burets shall start not less than 3 cm nor more than 7.5 cm from the upper end. The graduation marking for the 50 and 100 ml burets shall start not less than 4 cm nor more than 10 cm from the upper end.

3.4.5 Graduation marking.

3.4.5.1 Graduation marks.--All burets shall have graduation marks spaced so as to deliver at 20° C the quantity indicated by the numerical markings. Graduation markings shall be sharply defined lines of uniform width. Within the reading area line width shall not exceed 0.3 mm or vary more than 0.1 mm. Graduation markings shall be perpendicular to the vertical axis of the buret and parallel to each other. The distance between center line of adjacent marks of the buret shall be not less than 1 mm. The value of all main graduation markings shall be suitably inscribed in Arabic numerals directly above, but not touching, the line referenced. Graduation markings on burets shall be applied by one of the following methods--etched and filled with a permanent pigment, by application of a stain which is fired into the glass without etching, or by application of an enamel which is fused on the glass without etching. There shall be no appreciable change in the appearance of markings (see 3.4.5.2, 3.4.5.3, and 3.5) when tested as specified in 4.5.4.

3.4.5.2 Class A.--The least graduation markings shall extend a minimum of 1/2 and intermediate graduation markings shall extend approximately 3/4 (for convenience of reading), the way around the outside circumference of the buret. The main graduation markings, if etched and filled, shall extend completely around the outside of the buret. For stained or enameled lines, gap at closure shall not exceed 2 mm for 4 ml burets, 3 mm for 10 to 25 ml burets, and 4 mm for 50 to 100 ml burets, the gap position to be immediately to the right of numbers. The quality of graduation lines on class A burets shall be such that the method for setting the meniscus as outlined in NBS Circular 602 can be used. The setting arc is defined for full circumference and 3/4 circumference lines as starting at the ends of the short lines and extending into the clear area on the front face a distance of 1/8 of the circumference, and that portion of the back which is 180°. Within the setting arc, the bottom edge of the full (and 3/4 circumference) line on the front, and the top edge of this line on the back shall be straight. The opposite edges of the line; that is, the edges not used for reference in setting the meniscus, may show line width variation of 0.1 mm not to exceed 2 mm long. Etched lines shall be properly filled within the setting arc. Line or filler breaks which do not degrade a uniform appearance outside the setting arc are acceptable. Line or filler breaks which do not exceed 1 mm within the setting arc for the 25, 50 and 100 ml burets, and which do not exceed 0.5 mm on the 5 and 10 ml buret, are acceptable. The ends of the 1/2 circumference lines shall be free of defects on the bottom edge in the front and on the top edge in the back.

3.4.5.3 Class B.--The least graduation markings shall extend a minimum of 1/6 and intermediate graduation markings shall extend a minimum of 1/4 the way around the outside circumference of the buret. If the graduation markings are etched, the main graduation markings shall extend completely around the outside circumference of the buret; if the graduation markings are stained or enameled without etching, the main graduation markings shall extend a minimum of 3/4 the way around the outside circumference of the buret.

3.5 Identification Marking.--All burets shall be permanently and legibly marked with the manufacturer's name or trademark, the capacity, the delivery time, the temperature at which the graduations apply (20° C) and the method of use, that is to deliver. The delivery time inscription shall be at least 3 mm below the bottom graduation line. Each class A buret shall be marked with a serial number. Class B burets may be marked with serial number at the option of the manufacturer. Class A burets shall be further marked with the letter A to signify compliance with applicable construction and accuracy requirements.

3.6 Workmanship.--All burets shall be transparent and free from striae, surface irregularities, and other defects which may distort the appearance of the liquid surface or the portion of the graduation line seen through the glass.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures.--The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

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4.2.2 Supplemental evidence.--The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.2, 3.1.3, 3.4.2, and section 5 have been satisfied.

4.3 Lotting.--A lot shall consist of the burets of the same type, same style, same class, and same size produced by one manufacturer for delivery at one time. Buret lots shall be further defined by the type of stopcock plug required (see 6.2).

4.4 Sampling.

4.4.1 Sampling for visual examination.--Sampling for visual examination shall be in accordance with MIL-STD-105, Inspection Level II. and the classification of defects and AQL's indicated in table V.

Table V.--Classification of defects for visual examination.

Categories	Defects	Reference Paragraph
Class A		
Major:	AQL 1.5 percent defective	
	"A" mark missing.	3.5
	Marking missing or improper.	3.5
	Tubing not straight	3.4.1
	Stopcock not of proper type.	3.4.2
	Tips not properly ground and beveled or fire polished.	3.4.3
	Chips and cracks (other than obvious shipping damage).	3.6
	Smears of filler, scratches, acid burns, and other blemishes in reading area which interfere with setting.	3.6
	Graduation lines of wrong width or not straight. (Where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain.)	3.4.5.1
	The least, intermediate, or main graduation markings do not extend the distance specified to facilitate easy reading.	3.4.5.2
	Line or filler breaks degrade a uniform appearance outside the setting arc.	3.4.5.2
	Line or filler breaks in setting arc.	3.4.5.2
	Interval numbers or other inscription intersect graduation lines. (The interval numbers may touch the top of a graduation line, but not the end of a short line.)	3.4.5 3.5

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Table V.—Classification of defects for visual examination (continued)

Categories	Defects	Reference Paragraph
Class B		
Major:	AQL 4.0 percent defective	
	Marking missing or improper.	3.5
	Tubing not straight.	3.4.1
	Stopcock not of proper type.	3.4.2
	Tips not properly ground and beveled or fire polished.	3.4.3
	Chips and cracks (other than obvious shipping damage).	3.6
	Smears of filler, scratches, acid burns, and other blemishes in reading area which interfere with setting.	3.6
	Graduation lines of wrong width or not straight. (Where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain.)	3.4.5.1
	The least, intermediate, or main graduation markings do not extend the distance specified to facilitate easy reading.	3.4.5.3
	Interval numbers or other inscription intersect graduation lines. (The interval numbers may touch the top of a graduation line, but not the end of a short line.)	3.4.5.3 3.5

4.4.2 Sampling for tests.

4.4.2.1 Capacity accuracy, delivery time and stopcock leakage.--Sampling shall be in accordance with MIL-STD-105, Inspection Level S-4. For class A an AQL of 1.0 percent defective shall apply to the three tests combined. For class B an AQL of 2.5 percent defective shall apply to the three tests combined. (see 4.5.2 and 4.5.3).

4.4.2.2 Annealing and permanency of marking tests.--A sample of three burets shall be taken at random from each 100 burets or fraction thereof in the lot. Each buret in the sample shall be tested for annealing (see 4.5.1) prior to permanency of marking tests. A different one-third of the sample shall be used for tests as specified in 4.5.4.1, 4.5.4.2.1, and 4.5.4.2.2. If a buret fails to meet any test listed in this paragraph, the lot shall be rejected.

4.4.3 Preparation for delivery for classes A and B.--Defects listed in table VI shall be classified as major defects. Sampling shall be in accordance with MIL-STD-105, Inspection Level II, and an AQL of 2.5 percent defective.

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Table VI.--Classification of defects for preparation for delivery.

Unit, intermediate, or shipping container not as specified.
Number of burets per intermediate or shipping container not as specified or indicated.
Unlike burets packaged per intermediate, or shipping container not as specified.
Closure of unit, intermediate, or shipping container not as specified.
Unit, intermediate, or shipping container damaged.
Cushioning material not as specified.
Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.5 Tests.--Distilled water and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.5.1 Annealing.--Burets shall be tested in accordance with Method A or B of ASTM C 148.

4.5.2 Capacity accuracy.--Class A and class B burets shall be tested for capacity accuracy in accordance with the test procedure of section 3.5 of NBS Circular 602 or equivalent.

4.5.3 Stopcock leakage.--The buret shall show no leakage with the plug in either of the fully closed positions when filled to the zero line with water. Three-way stopcocks, in addition to complying with this test, shall show no leakage when tested similarly, with the buret empty, the plug in the normal delivery position, and the filling tube connected to a suitable graduated tube filled with water to a level 25 cm above the zero line of the buret.

4.5.4 Permanency of marking.--A different buret shall be used for each of the following tests:

4.5.4.1 Resistance to alkali.--The buret, with stopcock removed, shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.5.4.2.1 Resistance to acid.

4.5.4.2.1 Hydrochloric acid.--The buret shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing rigorously with a cloth.

4.5.4.2.2 Sulfuric acid.--The buret shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging.--Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging.--Each buret shall be packaged in box of appropriate size conforming to PPP-B-566, style I or II, type and class optional or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closure shall be secured with tape conforming to UU-T-111.

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5.1.1.2 Intermediate packaging.--The number of burets specified (see 6.2), of the same type, class, and size, packaged in accordance with 5.1.1.1, shall be intermediately packaged in a snug-fitting fiber box conforming to PPP-P-636, type CF, class domestic. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C.--Burets shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2. Packing.--Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.--Burets shall be packed in a box conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification and appendix thereto.

5.2.2 Level B.--Burets shall be packed in fiberboard boxes conforming to PPP-B-636, type CF, class domestic, style RSC, minimum bursting strength 275 pounds per square inch.

5.2.3 Level C.--Burets shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies.--In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies.--In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.--Burets covered by this specification are intended for laboratory applications. In general, class B burets will be most applicable for Federal and Military agencies. Class A burets are intended for precision measurements and are of a quality such that specific burets will be acceptable by the National Bureau of Standards for calibration at specified points if such calibration is justified by the nature of the usage.

6.2 Ordering data.--Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Type, style, class and size required.
(d) Type of stopcock plugs.
(e) Number of burets per intermediate package required.

6.3 Military use.--Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards or military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Transportation description.--Transportation description applicable to this item is:

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds, subject to Rule 34,
Uniform Freight Classification.
Glassware, laboratory.
Motor volume minimum weight 18,000 pounds.

MILITARY INTERESTS:

Army - MU
Navy - MS
Air Force - 67

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